Sreevaatsav Bavana

Hyderabad, Telangana, India · *Portfolio:* ***sreevaatsavb.github.io/portfolio***

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Personal Profile

Gold medalist in B.Tech in Artificial Intelligence at Mahindra University; currently a Senior Data Scientist at Penguin AI. Actively learning and working on robotic learning with a background in multimodal AI. Research interests include long-horizon task planning for contact-rich manipulation and reinforcement learning (RL) post-training of robot policies.

Education

Mahindra University, Hyderabad, India *Aug 2020 - Aug 2024*

Bachelor of Technology in Artificial Intelligence

- **Gold Medalist** of the 2020–2024 B.Tech AI batch
- **Cumulative GPA** : 8.76
- **Relevant Courses:** Introduction to Robotics, Control Theory, Reinforcement Learning and Autonomous Systems, Signals & Systems, Neural Networks and Deep Learning, Machine Learning and Python, Optimization Techniques for AI, Data Structures

Publications

GIV-CXR: Densely Grounded, Visually Interpretable Chest X-ray Question Answering Dataset *2026*

Sreevaatsav Bavana, Adit Rushil Potta, Sai Amrit Patnaik, Nidhi Goyal

Workshop on Medical Reasoning with Vision Language Foundation Models (MedReasoner), ***CVPR 2026****, IEEE/CVF*

From Layoff to Reemployment: Analyzing Social Media Narratives *2025*

Yayati Gupta, Sreevaatsav Bavana, Sudhira Chegu, Sujith Sai Kalakonda, Yash Mathur, Manmitha Polsani, Sanatan Sukhija

International Conference on Computational Data and Social Networks, ***CSoNet 2025****, Springer LNCS*

A Systematic Study on Video Summarization: Approaches, Challenges, and Future Directions *2023*

Kajal Kansal, Nikita Kansal, Sreevaatsav Bavana, Bodla Krishna Vamshi, Nidhi Goyal

Workshop on User-centric Narrative Summarization of Long Videos (NarSUM), ***ACM MM 2023****, ACM*

DOI: 10.1145/3607540.3617139

Projects

Imitation-Learning Policies on a Fairino FR5 Cobot *Mar 2026 - Present*

Sliact Robotics (stealth startup), Founding Engineer · **Write-up: 3-part blog series**

- Collected 1,021 teleoperated demonstrations (∼11.5 hours) on a 6-DOF Fairino FR5 for colour sorting, sanding and pick-to-light, first with an SO-101 leader arm and later a Meta Quest controller over ROS 2. Synced the 100 Hz robot logs with 30 Hz frames from a wrist-mounted RealSense D405 and a scene camera.
- Trained ACT, Diffusion Policy, Octo, π_0 and $\pi_{0.5}$ and ran them closed-loop on the arm at 30 Hz. Fully fine-tuned π_0 on an A100, starting the sanding run from the colour-sorting checkpoint.
- π_0 succeeded in 80% of sanding and 65% of colour-sorting trials (20 autonomous runs each). XR-1 adaptation is in progress.
- Connected π_0 to a VLM planner (Sonnet 5.5) for multi-step tasks and found that neither could identify contact issues. Now adding a force/torque sensor and trying RL post-training with human corrections for sanding and screwing.

AnimalCLEF 2026: Wildlife Re-identification (9th Place) *Mar 2026 - May 2026*

AnimalCLEF 2026 Challenge

- Grouped ∼2,400 unlabeled images by individual animal across 4 species (lynx, salamanders, sea turtles, Texas horned lizards).
- Combined global embeddings from MiewID v3 and MegaDescriptor with local feature matching (KAZE, ALIKED + LightGlue).
- Fine-tuned MegaDescriptor with ArcFace loss (+6.4%) and used SAM2 to mask out background keypoints before matching.

GIV-CXR: Densely Grounded, Visually Interpretable Chest X-ray Question Answering Dataset *Aug 2024 - Jan 2025*

VLaNC Lab, Mahindra University (Dr. Nidhi Goyal) · in collaboration with IIIT Hyderabad

- Built a chest X-ray question-answering benchmark with 355,293 question-answer pairs, each tied to bounding boxes across 36 anatomical regions, to test spatial reasoning.
- Set up a hallucination filter that combines automatic checks with radiologist review, with an 82.4% acceptance rate.
- Trained grounded VLM baselines. The best one reached 68.1 mIoU on spatial grounding, nearly double GPT-4o-mini's 39.0, and the top G-Eval answer score (3.98).

Query-Based Video Summarization with a Learnable Text-to-Frame Projection *Aug 2023 - Dec 2023*

Supervisor: Dr. Nidhi Goyal, Assistant Professor, Mahindra University

- Added a learnable GELU projection that maps CLIP text-query embeddings into the video-frame embedding space of Moment-DETR, a transformer that finds the moments and highlights in a video that match a text query.
- Trained it on a subset of QVHighlights and improved mAP by ∼18% over the original Moment-DETR.

Work Experience

Senior Data Scientist *May 2026 - Present*

Penguin AI, Hyderabad, India

- Lead the AI module of a prior-authorization agent system for a large US health plan (LangGraph, AWS Bedrock) that turns clinical guidelines into questionnaire trees, judges each atomic criterion as met or not met from patient records, and rolls the answers up to a decision. Processed 18,500+ cases across 110 guidelines with 99.4% pipeline success and a 4.4-minute median latency.
- Built a guideline selector based on self-consistency sampling that corrected 830 of 1,185 audited routing decisions. With this and other accuracy fixes, agreement between the AI and human reviewers rose from 31% to 67% per procedure line in four months (Orthopedics: 47% to 77%), and unsafe approvals stayed under 1%.
- Helped raise turnaround-time compliance from 44% to 92% and cut reviewer time per line by nearly 40% (4.4 to 2.8 min) across 41 clinical reviewers.

Data Scientist *Jul 2025 - May 2026*

Penguin AI, Hyderabad, India

- Built a claim-denial prediction system end to end, from per-client ETL to a LightGBM and XGBoost ensemble trained with MLflow, with PSI drift alerts and periodic retraining. It reached 0.87 AUC and caught 92% of denials on 18,823 held-out claims.
- Built an LLM agentic workflow that turns payer medical-policy PDFs into questionnaire trees. It scored higher question F1 than the existing drafts (0.64 vs 0.60), at $0.30 to $0.60 and 5 to 10 minutes per policy.

Associate Data Scientist *Jul 2024 - Jun 2025*

Awone AI, Hyderabad, India

- Made production LLM inference 15% faster with quantization and an n-gram speculative decoding head.
- Built a medical PII redaction system that combined a BERT NER model, trained with a custom loss, with document vision-language models. It redacted correctly 90% of the time, against 60-70% for standard tools.
- Trained document vision-language models to fill in forms automatically, with 82% accuracy on forms they had not seen before.
- Built a voice chat app from open-source components that users can interrupt mid-reply, plus an MCP client and server for querying a user database.

Data Science Intern *Jan 2024 - Jun 2024*

Awone AI, Hyderabad, India

- Re-implemented Llama-2, Mistral, Qwen2 and LLaVA from scratch to understand their architectures and attention.
- Explored generative and multimodal models, including diffusion models built from scratch, Stable Diffusion 3, speech LLMs that output audio, and PaliGemma.
- Studied ways to speed up LLM inference, such as speculative decoding with n-gram and Medusa heads, KV caching, FlashAttention and quantization.

Summer Research Intern *Jul 2023 - Sept 2023*

Mahindra University, Hyderabad, India (Supervisor: Dr. Nidhi Goyal)

- Surveyed video summarization methods with a four-person team, in collaboration with a researcher from the National University of Singapore (NUS). The survey was published at the NarSUM workshop, ACM MM 2023.

Achievements

9th place in AnimalCLEF 2026 Wildlife Re-identification Challenge *2026*

Merit scholarship for all four years (2020-2024) for ranking in the top 5% of the branch *2024*

3rd place in ICETCI Hackathon 2023, Mahindra University *2023*

Rank 111 of 4,000+ teams in Amazon ML Challenge 2023 *2023*

Courses & Certifications

Modern Robotics, Course 1: Foundations of Robot Motion *Apr 2026*

Coursera, Northwestern University · Grade: 98.46%

Modern Robotics, Course 2: Robot Kinematics *In Progress*

Coursera, Northwestern University

Skills

Languages & Libraries	Python, SQL, Pandas, NumPy, Scikit-learn, HuggingFace, PyTorch, OpenCV, C++
AI Engineering & Infrastructure	LangChain, LangGraph, vLLM, ONNX, MLFlow, Ray
Cloud & Deployment	AWS, Docker, Async programming, Linux, GitHub Actions, MongoDB, PostgreSQL, Kubernetes
Robotics & Simulation	ROS 2, Robot kinematics, MuJoCo, CoppeliaSim